

JavaScript Array Methods — Realistic Practice Scenarios

Using this array:

```
var studentMarks = [  
  {  
    subject: "English",  
    marks: 90,  
    teacher: "Ramesh",  
    pass: true,  
    grade: "A",  
    totalMarks: 100  
  },  
  {  
    subject: "Maths",  
    marks: 95,  
    teacher: "Suresh",  
    pass: true,  
    grade: "A+",  
    totalMarks: 100  
  },  
  {  
    subject: "Science",  
    marks: 85,  
    teacher: "Priya",  
    pass: true,  
    grade: "B+",  
    totalMarks: 100  
  },  
  {  
    subject: "Social",  
    marks: 90,  
    teacher: "Anitha",  
    pass: true,  
    grade: "A",  
    totalMarks: 100  
  },  
  {  
    subject: "Computers",  
    marks: 75,  
    teacher: "Kiran",  
    pass: true,  
    grade: "B",  
    totalMarks: 100  
  }  
]
```

```
}  
];
```

1. map()

Scenario 1

School management wants only subject names.

Expected Output

```
["English", "Maths", "Science", "Social", "Computers"]
```

Scenario 2

Add 5 grace marks to every subject.

Scenario 3

Convert all subject names to uppercase.

2. filter()

Scenario 1

Find subjects with marks greater than 90.

Scenario 2

Find subjects handled by teacher "Priya".

Scenario 3

Find all subjects having grade "A".

3. find()

Scenario 1

Find the first subject with marks below 80.

Scenario 2

Find the subject taught by "Suresh".

4. some()

Scenario 1

Check whether any subject has failed.

Scenario 2

Check whether any subject has marks below 35.

5. every()

Scenario 1

Check whether all subjects are passed.

Scenario 2

Check whether all subjects scored above 70.

6. reduce()

Scenario 1

Calculate total marks scored by the student.

Scenario 2

Find average marks.

Scenario 3

Find highest marks.

7. sort()

Scenario 1

Sort subjects by marks ascending.

Scenario 2

Sort subjects by marks descending.

Scenario 3

Sort subjects alphabetically.

8. forEach()

Scenario 1

Print all subject names with marks.

Example Output

```
English - 90  
Maths - 95
```

9. includes()

Scenario 1

Check whether "Maths" subject exists.

Hint: Extract subjects first.

10. Chained Methods

Scenario 1

Get only subject names where marks are above 90.

Expected Output

```
[ "Maths" ]
```

Scenario 2

Find average marks of only grade "A" subjects.

11. Realistic Dashboard Tasks

Scenario

School dashboard needs:

- Total subjects
 - Passed subjects count
 - Failed subjects count
 - Highest scoring subject
 - Lowest scoring subject
-

12. Advanced Thinking Practice

Scenario 1

Convert array into this object:

```
{  
  English: 90,  
  Maths: 95,  
  Science: 85  
}
```

Scenario 2

Group subjects by grades.

Expected Output

```
{  
  A: ["English", "Social"],  
  "A+": ["Maths"],  
  "B+": ["Science"]  
}
```

Best Order to Practice

1. forEach
2. map
3. filter
4. find
5. some
6. every
7. sort
8. reduce
9. chaining methods
10. advanced transformations

Important Thinking Pattern

Question	Method
Transform every item?	map
Select some items?	filter
Find one item?	find
Check condition on any item?	some
Check condition on all items?	every

Question	Method
Calculate one final value?	reduce
Reorder items?	sort

This thinking pattern is what experienced JavaScript developers use daily.